ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

ELEC2134 : Circuits and Signals (PART B)

Topic 6a: AC Circuit Analysis

Alternating Current (AC)

- ✓ AC current and voltage
- ✓ Phase angle and Phase difference
- ✓ Peak and peak-peak values
- ✓ Average value
- ✓ Root Mean Square (RMS) value

Alternating Current (AC)

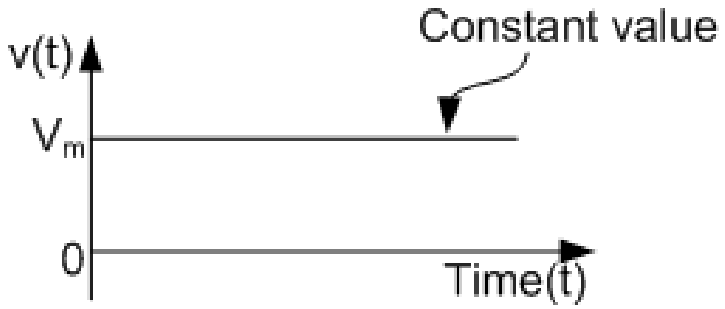
- ✓ Phasor Representation and Phasor diagrams
- ✓ V-I relationship for resistor/capacitor/ Inductor
- ✓ Impedance and reactance
- ✓ Analysis of parallel RC/ RL circuits
- ✓ Analysis of RLC circuits

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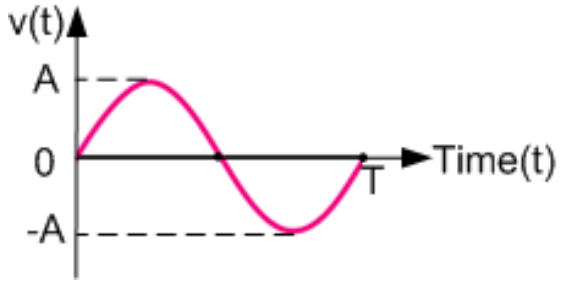
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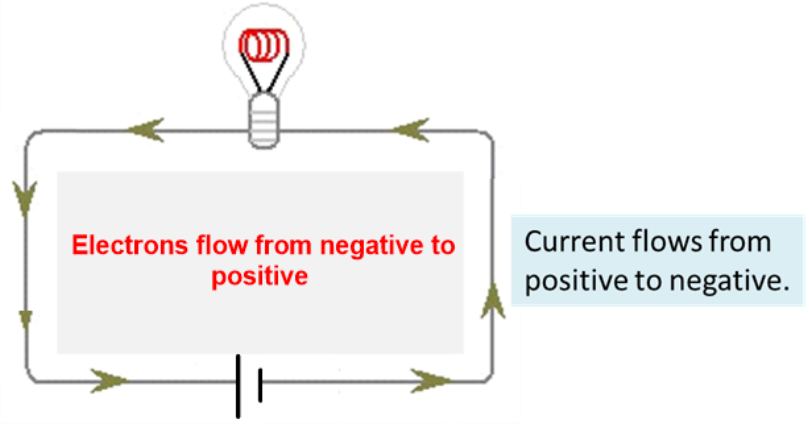
AC Fundamentals

- In Direct Current (DC), the electric charge flows only in one direction.

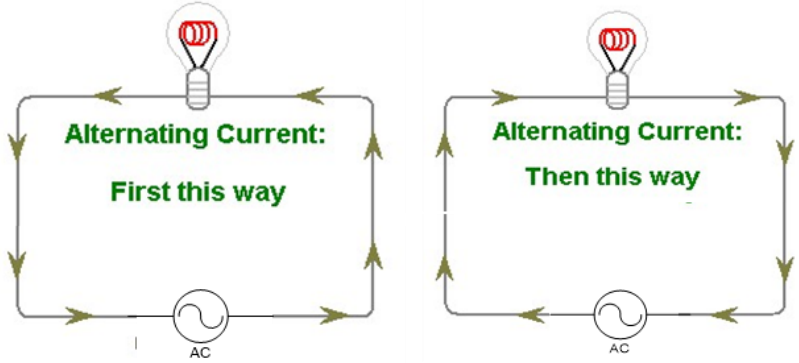


- In Alternating Current (AC), the flow of electric charge periodically reverses direction.





DC power sources:
Batteries, Solar cells



AC power sources:
Generators in power plants, wind turbines etc.

AC Transmission

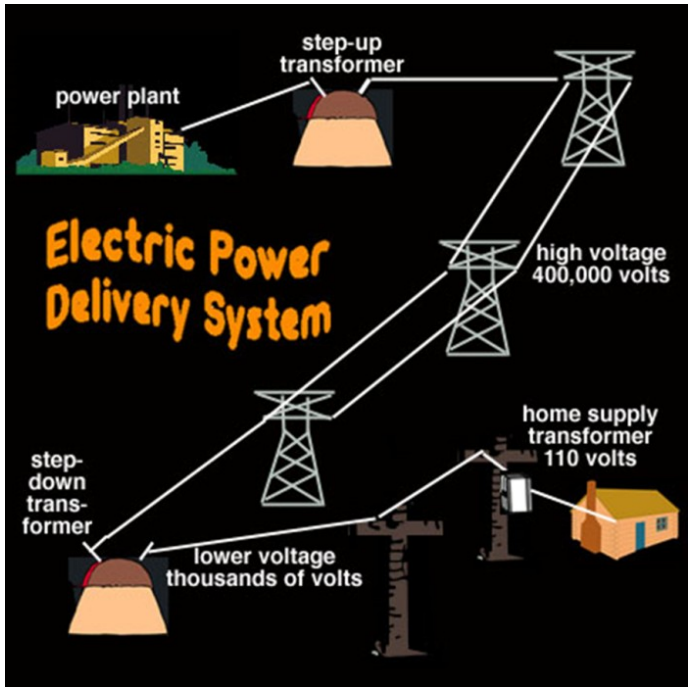
- AC is more efficient and economical to transmit over long distances
- AC voltage may be increased or decreased with a transformer
- The power losses (P_L) in a conductor are a product of the square of the current and the resistance of the conductor

$$\text{Power loss} = I^2 R$$

- When transmitting a fixed power through a power line, if the current is doubled, the power loss will be four times greater.
- The power transmitted (P_T) is equal to the product of the current and the voltage

$$P_T = V\,I$$

- The same amount of power can be transmitted with a lower current by increasing the voltage
- Use of a higher voltage leads to significantly more efficient transmission of power.



Source : http://www.windows2universe.org

Note: DC cannot be step-up or step- down by a transformer

Note: High voltage DC transmission (HVDC) is becoming available but tend to be more expensive less efficient over short distances.

Characteristics of Alternating Current and Voltage (AC)

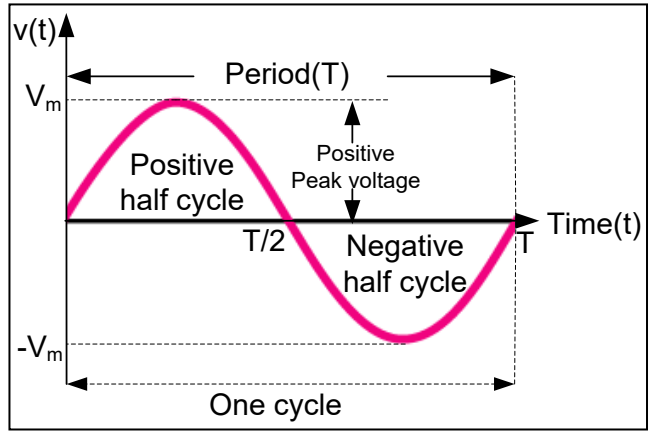
- An alternating current or voltage can have different waveforms, but we will focus only on the sine wave (sinusoidal waveform).

$$\boldsymbol{v(t) = V_m\, sin(2\pi f t)}$$

Instantaneous voltage — Peak voltage — frequency — time

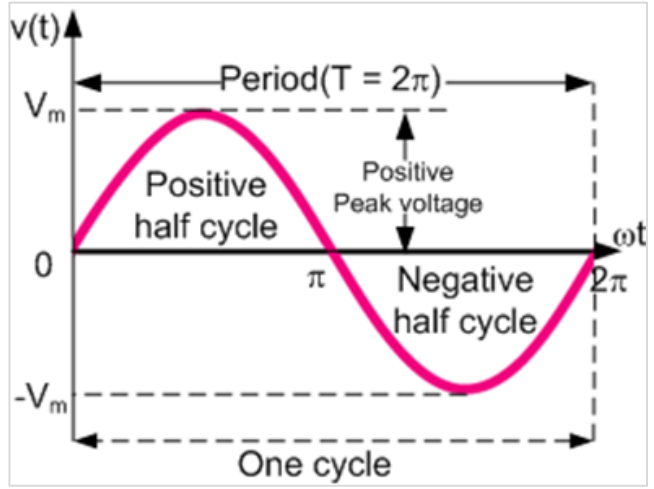
$$v(t) = V_m sin(\frac{2\pi}{T} t)$$

- Therefore $v(t)$ can be re-written as $\boldsymbol{v(t) = Vm\, sin(\omega t)}$



Frequency(f)= $\frac{1}{T}\ Hz\ (Hertz)$

Angular Velocity ($\boldsymbol{\omega}$)=$\boldsymbol{2\pi f = \frac{2\pi}{T}\ radians/sec}$

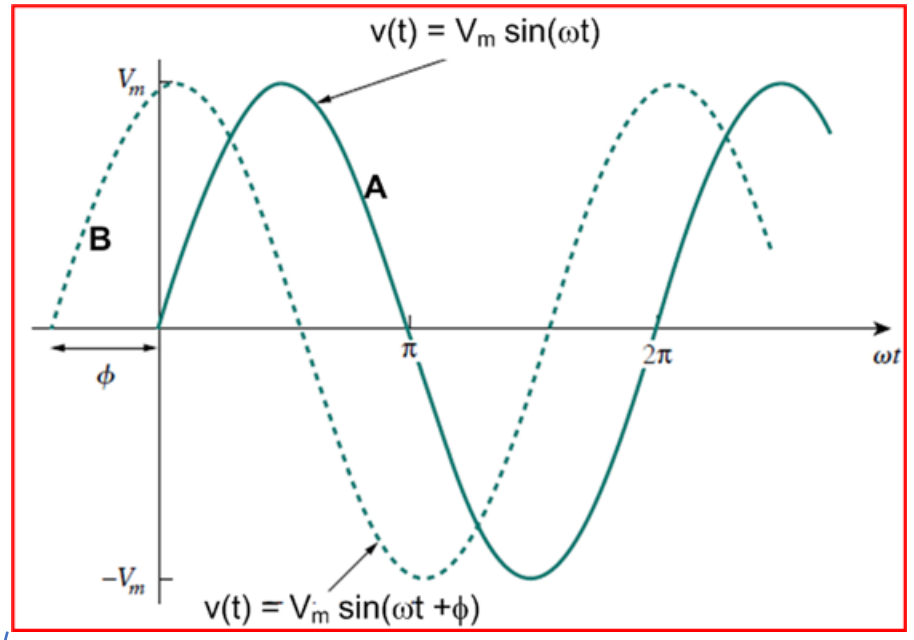


Phase Angle

- A sinusoidal waveform may or may not always start at $t = 0$. It can start earlier or later than $t = 0$.
- A **delay** or **advance** in the waveform is known as the **phase angle** (ϕ). The units of ϕ is given in degrees or radians.
- A general expression for a sinusoidal waveform is given by

$$v(t) = Vm\ sin(\omega t \pm \emptyset)$$

- Note that the phase angle ϕ can take any values between 0 to 360 degrees



- Two sinusoids (see above diagram) with a phase difference
- Waveform "*A*" is chosen as the reference and is shown to begin at 0°.
- Waveform "*B*" **leads** waveform "*A*" by ϕ degrees. Therefore it is written as
- $$v(t) = V_m sin(\omega t + \emptyset)$$

Note: If ϕ is not zero, the *waveforms A and B* are said to be out of phase. If $\phi = 0$, then the *waveforms A and B* are said to be in phase

Phase Difference

- Waveform A is chosen as the reference and is shown to begin at 0°.

$$v(t) = V_m \sin(\omega t)$$

- Waveform B leads waveform A by 45 degrees

$$v(t) = V_m \sin(\omega t + 45^\circ)$$

- Waveform C lags waveform A by 60 degrees

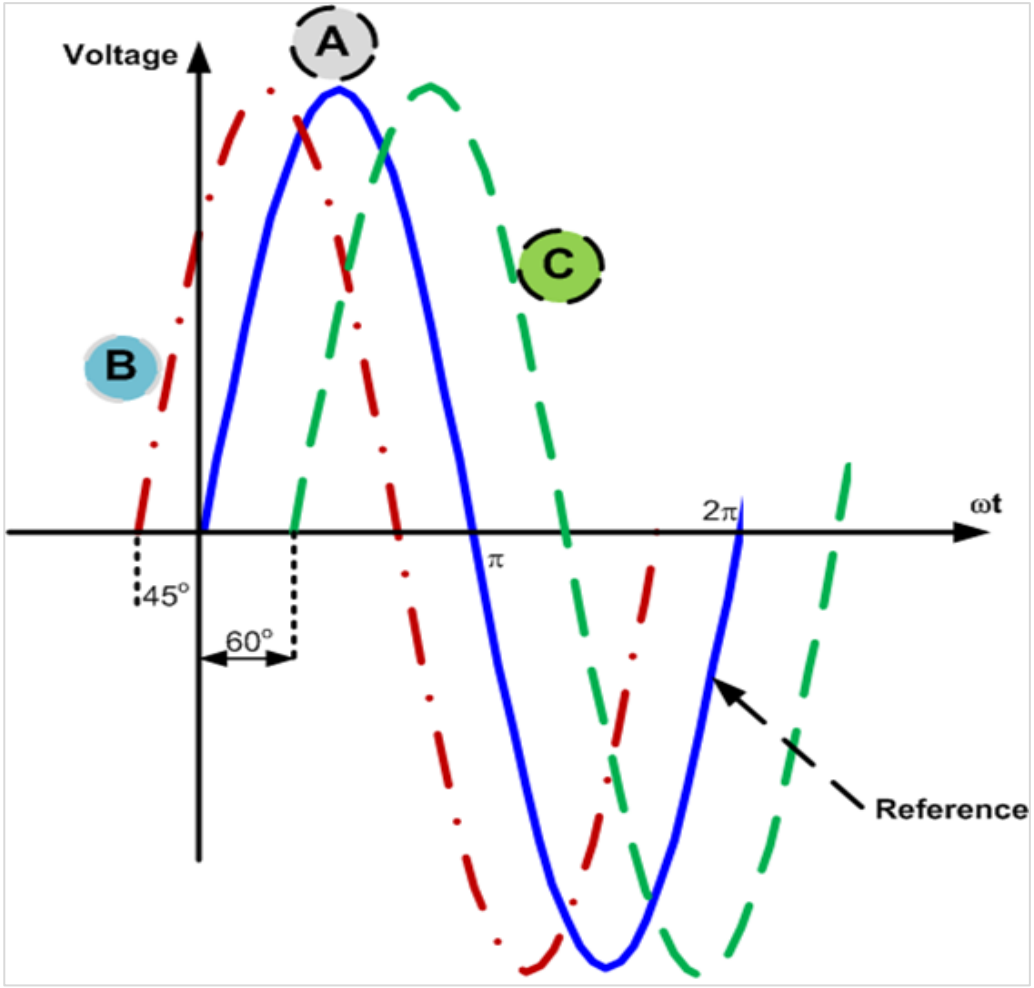
$$v(t) = V_m \sin(\omega t - 60^\circ)$$

- **Note:** A sinusoid can also be expressed as a cosine function.

$$\boldsymbol{v(t) = V_m \cos(\omega t \pm \phi)}$$

$$v(t) = V_m \sin(\omega t + 90^\circ) = V_m \cos(\omega t)$$

$$v(t) = V_m \cos(\omega t - 90^\circ) = V_m \sin(\omega t)$$



Note: Waveform B leads waveform C by 105 degrees

Examples

Example 1

❑ Find the frequency *f*, period *T* of the voltage waveforms given below. Also find the phase angle between $v_1(t)$ and $v_2(t)$. Does $v_1(t)$ lead or lag $v_2(t)$?

$$(i) \quad v_1(t) = 12\sin(314t + 45^\circ) \text{ and } v_2(t) = 6\sin(314t - 15^\circ)$$

$$(ii) \quad v_1(t) = 12\sin(314t + 45^\circ) \text{ and } v_2(t) = 6\cos(314t - 105^\circ)$$

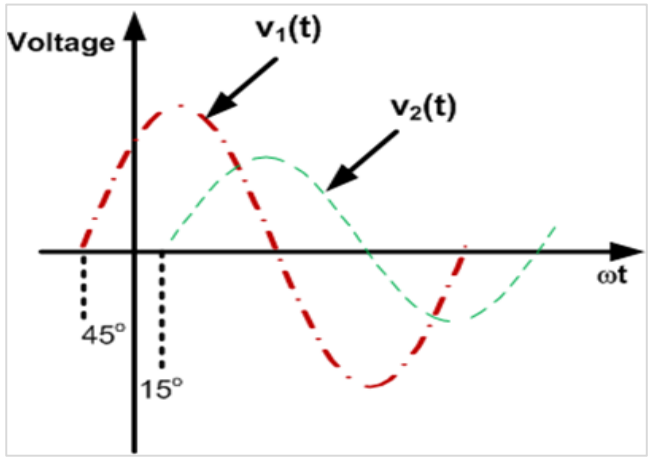
Solution:

(ι) $\omega t = 2\pi ft = 314t$

$f = 314/2\pi = 50$ Hz

T = 1/f =1/50 Hz

T= 0.02 sec = 20 ms



$v_1(t)$ **leads** $v_2(t)$ by 60 degrees.

(ii) $v_2(t) = 6\cos(314t - 105^\circ) = 6\cos[(314t - 15^\circ) - 90^\circ] = 6\sin(314t - 15^\circ)$

This is similar to part (i) above. So v_1 leads v_2 by 60 degrees.

Examples

Example 2

❑ Find the phase angle between $i_1 = -4\sin(100\pi t + 25^\circ)$ and $i_2 = 5\cos(100\pi t - 40^\circ)$ .Does i_1 lead or lag i_2?

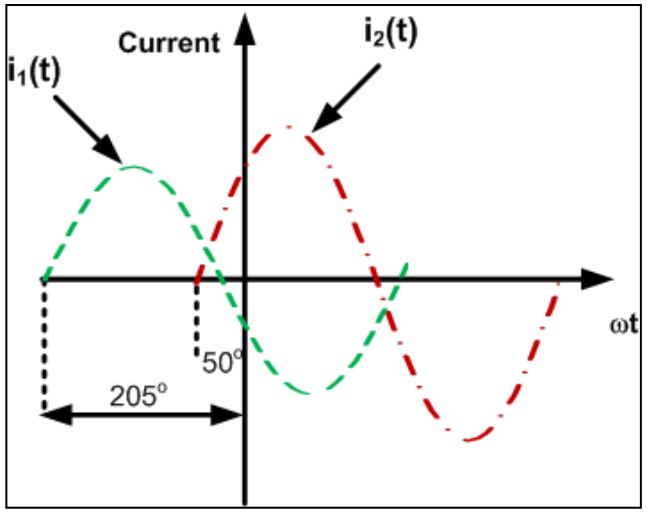
Solution:

$$i_1 = -4\sin(100\pi t + 25^o) = 4\sin(100\pi t + 25^o + 180^o) = 4\sin(100\pi t + 205^o)$$

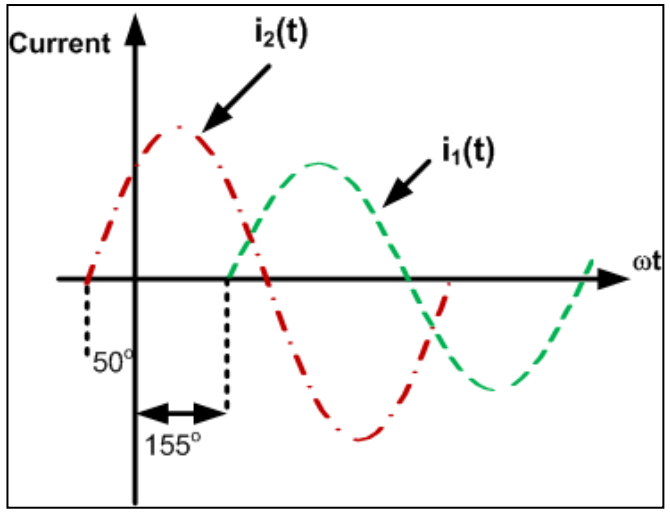
OR

$$i_1 = -4\sin(100\pi t + 25^o) = 4\sin(100\pi t + 25^o - 180^o) = 4\sin(100\pi t - 155^o)$$

$$i_2 = 5\cos(100\pi t - 40^o) = 5\sin(100\pi t - 40^o + 90^o) = 5\sin(100\pi t + 50^o)$$



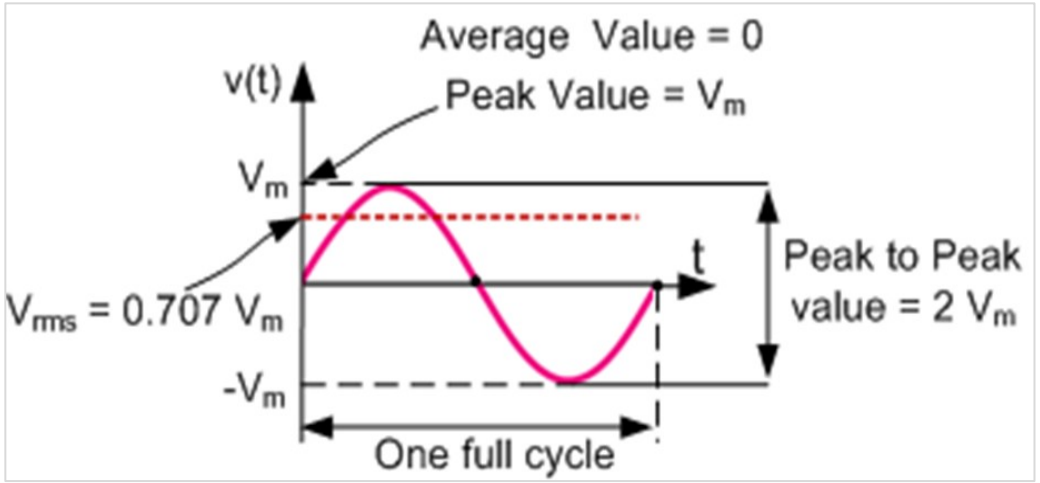
Therefore i_1 leads i_2 by 155 degrees.



Therefore i_1 lags i_2 by 205 degrees.

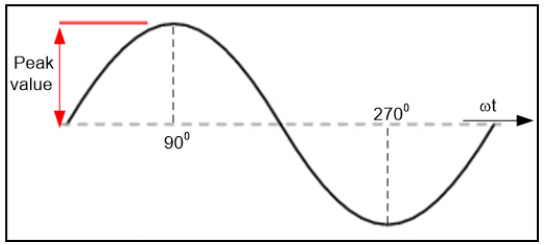
Measurements of AC Quantities

- ✓ From a sinusoidal waveform we can measure the following values:
  - Peak value
  - Peak to peak value
  - Average value
  - Root Mean Square (RMS) value



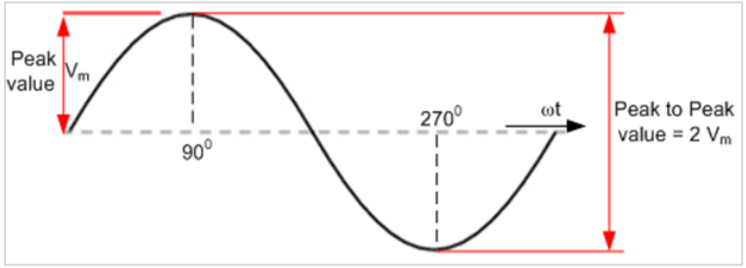
Peak Value

- The peak value of a sine wave is known as the maximum voltage (or current) of the waveform.
- Peak voltage (or current) occurs at two different positions in a cycle (one positive peak and one negative peak).
- The positive and negative peaks have equal magnitudes.
- The positive peak occurs at 90° and negative peak occurs at 270°.

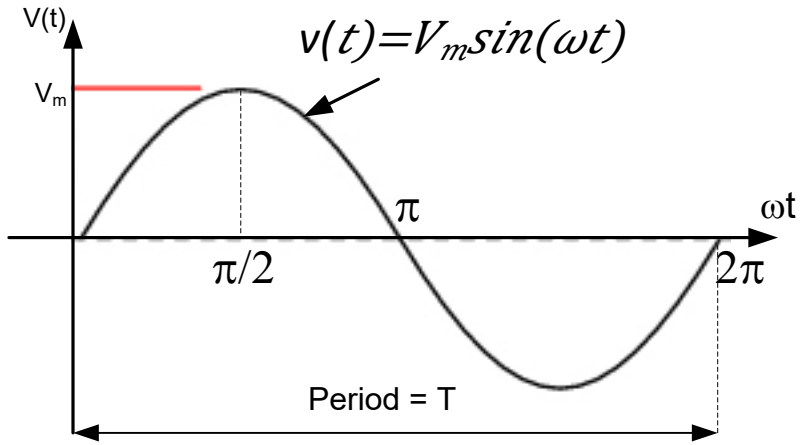


Peak to Peak Value

- The peak-to-peak (pk-pk) value of an AC waveform is the total height between opposite peaks.



Average Value



$$V_{ave} = \frac{1}{T}\int_0^T v(t)dt \qquad \omega = \frac{2\pi}{T}$$

$$= \frac{1}{T}\int_0^{2\pi/\omega} V_m \sin(\omega t)dt$$

$$= \frac{V_m}{T}\left[\frac{-\cos(\omega t)}{\omega}\right]_0^{\frac{2\pi}{\omega}}$$

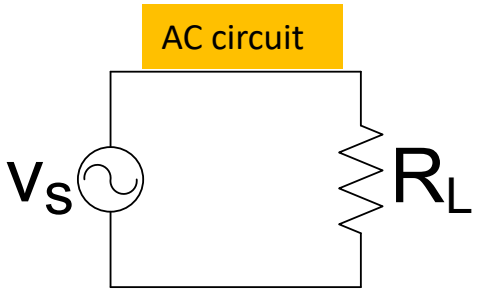
$$V_{ave} = \frac{V_m}{T\omega}[\cos(\omega t)]_{\frac{2\pi}{\omega}}^{0} = 0$$

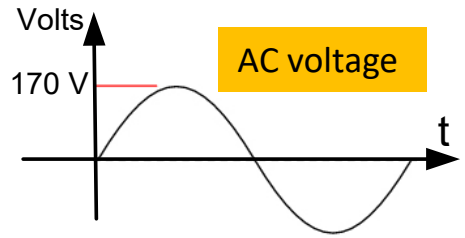
- ✓ **Average value** V_{ave} or I_{ave} of a sine wave is the average of all its instantaneous values over a full **cycle.**
- ✓ $V_{ave} = \frac{1}{T}\int_0^T v(t)dt = \frac{Area\ under\ the\ curve\ over\ a\ full\ cycle}{Period\ (T)}$
- ✓ Since the positive and negative alternations are identical, the average is zero.
- ✓ Thus, the average value of a sine wave, taken over an entire cycle is zero.

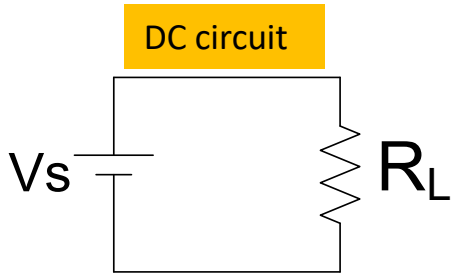
As expected the net area of the positive and negative half cycles is zero.

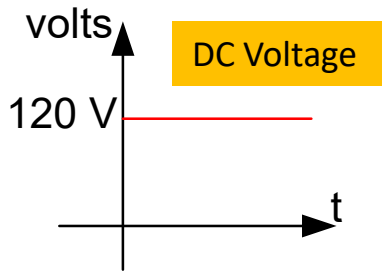
Root Mean Square Value (RMS)

- The root mean square value or **rms** value is actually a measure of the heating effect of a sine wave.
- It can be defined as the equivalent value of DC that will produce the same amount of heat (power) in a load (R_L) as a given sine wave.









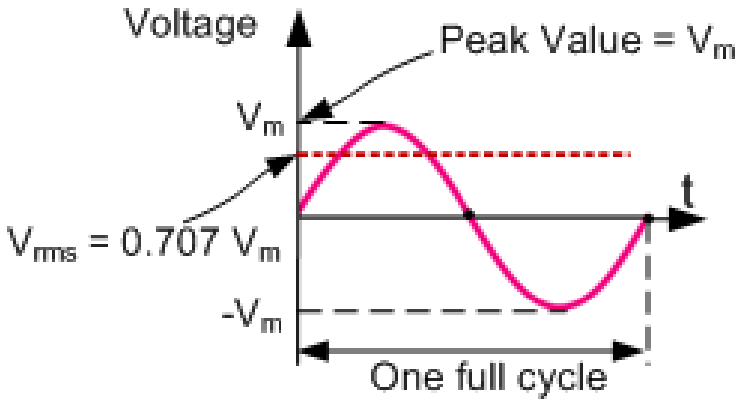
- A sine wave with a peak value of 170V (= 120 V RMS) is applied across a resistor (R_L), a certain amount of heat is produced.
- If a DC voltage with a steady-state value of 120 V is applied to the resistor (R_L), the same amount of heat is produced.

Note: When rms value of the sinusoidal source voltage is equal to the DC source voltage, then the AC source will supply the same average power as the DC source.

RMS Value of a Sinusoid

✓ We will show that the **rms** value of a sine wave of voltage or current is :

$$v_{rms} = \frac{V_m}{\sqrt{2}} \qquad \text{and} \qquad i_{rms} = \frac{I_m}{\sqrt{2}}$$



$$v(t) = V_m \sin(\omega t) = V_{rms}\sqrt{2}\,\sin(\omega t)$$

✓ Then the RMS voltage (V_{RMS}) of a sinusoidal waveform is determined by multiplying the peak voltage value by **0.707** (equivalent to dividing it by the square root of two)

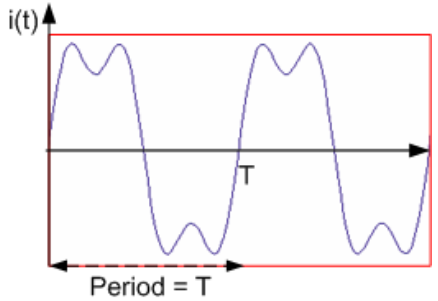
✓ The RMS voltage or current, which is also known as the effective value, of a sinusoid depends on the Amplitude of the waveform and is **independent** of the waveform **frequency its phase angle**.

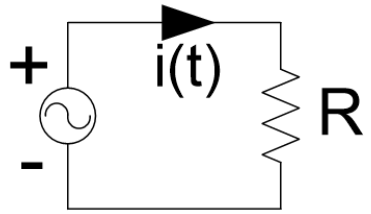
RMS value or Effective Value of a periodic Waveform

- The I_{rms} of a periodic current waveform *i*(*t*) is defined as the current which produces heat (power) in a given resistor R at the same average rate as a direct current I_{dc}.

$$P_{average} = R.I_{rms}^2 = RI_{dc}^2$$

✓ Consider a periodic current waveform





i(*t*) – instantaneous current
p(*t*) – instantaneous power
$p(t) = i^2(t)R$

$$P_{ave} = \frac{1}{T}\int_0^T p(t)dt = \frac{1}{T}\int_0^T i^2(t)R\,dt$$

$$P_{ave} = \frac{R}{T}\int_0^T i^2(t)dt = RI_{rms}^2$$

$$\therefore I_{rms}^2 = \frac{1}{T}\int_0^T i^2(t)\,dt \rightarrow I_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2(t)dt}$$

Root Square

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt}$$

Mean

RMS current calculation for a sinusoid

$$I^2_{rms} = \frac{1}{T}\int_0^T i^2(t)dt$$

We have T= $2\pi/\omega$

$$= \frac{\omega}{2\pi}\int_0^{\frac{2\pi}{\omega}} I_m^2 \sin^2(\omega t)dt$$

$$= \frac{\omega I_m^2}{2\pi}\int_0^{\frac{2\pi}{\omega}} \sin^2(\omega t)dt$$

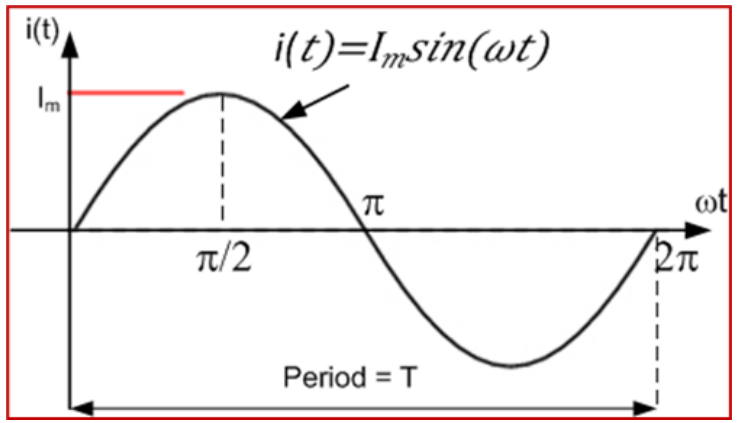
$$\cos(2\omega t) = 1 - 2\sin^2(\omega t)$$

$$= \frac{I_m^2\omega}{2\pi}\,\frac{1}{2}\int_0^{\frac{2\pi}{\omega}} 1 - \cos(2\omega t)\,dt$$

$$= \frac{I_m^2\omega}{2\pi}\,\frac{1}{2}\left\{\{t\}_0^{\frac{2\pi}{\omega}} - \left[\frac{\sin(2\omega t)}{2\omega}\right]_0^{\frac{2\pi}{\omega}}\right\}$$

$$= \frac{I_m^2\omega}{2\pi}\,\frac{1}{2}\,\frac{2\pi}{\omega} = \frac{I_m^2}{2}$$

$$\therefore \boldsymbol{I_{rms} = \frac{I_m}{\sqrt{2}}}$$



Note: We usually quote most AC voltages as RMS rather than peak value. If V_{rms} is 240 V, then

$$V_m = V_{rms}\sqrt{2}\ \text{volts}$$
$$= 240\sqrt{2}\ \text{volts}$$
$$= 339.5\ \text{volts}$$

Note:

- Consider a waveform consists of a sum of sinusoids of different frequencies.
- The rms value is equal to the square root of the sum of the squares of the rms values of each sinusoid.
- Assume that the ac current $i(t)$ is given by:

$$i(t) = I_0 + I_{m1}sin(\omega_1 t) + I_{m2}sin(\omega_2 t) + \ldots\ldots$$

Therefore

$$I^2_{rms} = I_0^2 + I^2_{rms1} + I^2_{rms2} + \ldots.$$

Note: UK: 220 V_{rms}; USA: 110 V_{rms} ; Australia: 240 V_{rms};

ELEC2134 : Circuits and Signals (PART B)

Topic 6: AC Circuit Analysis

Alternating Current (AC)

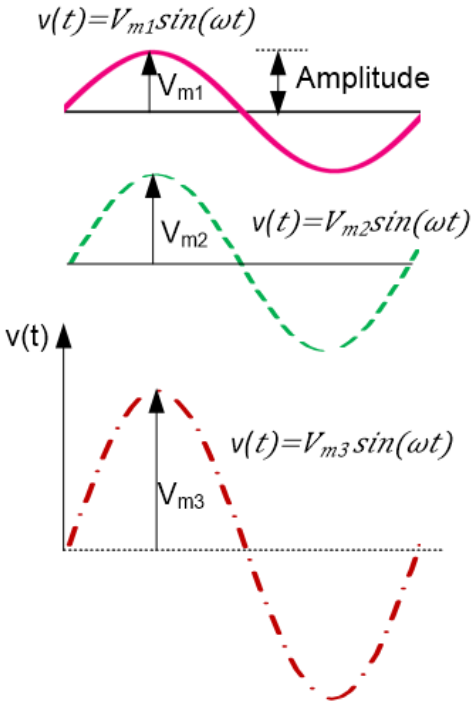
- ✓ Phasor Representation and Phasor diagrams
- ✓ V-I relationship for resistor/capacitor/ Inductor
- ✓ Impedance and reactance
- ✓ Analysis of parallel RC/ RL circuits
- ✓ Analysis of RLC circuits

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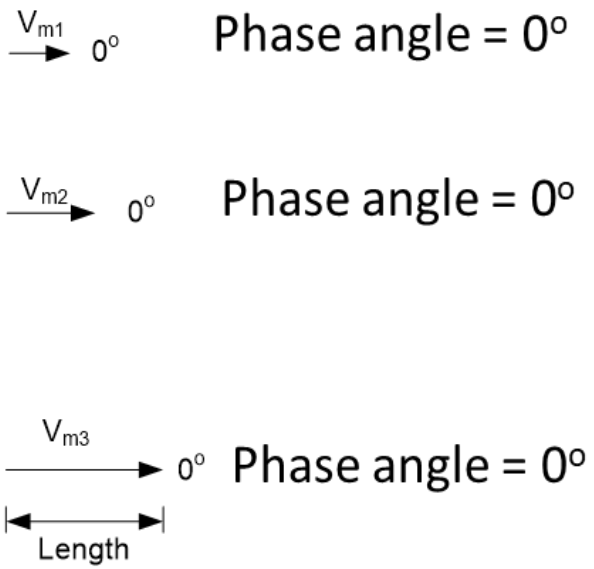
Phasor Representation

- A phasor is a line , drawn from an origin that has both magnitude and angle
- Phasors are used in AC circuits to indicate time varying quantities of current and voltage

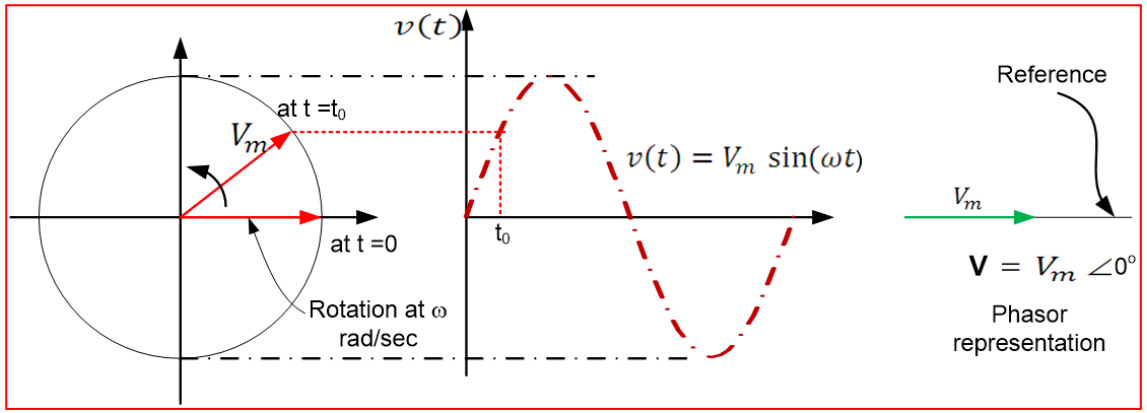
Waveform

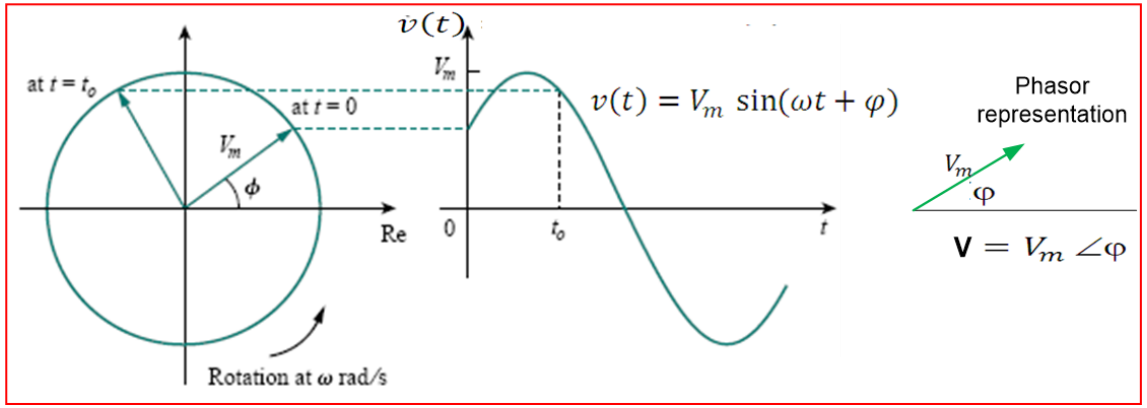


Vector representation



A phasor is a rotating vector

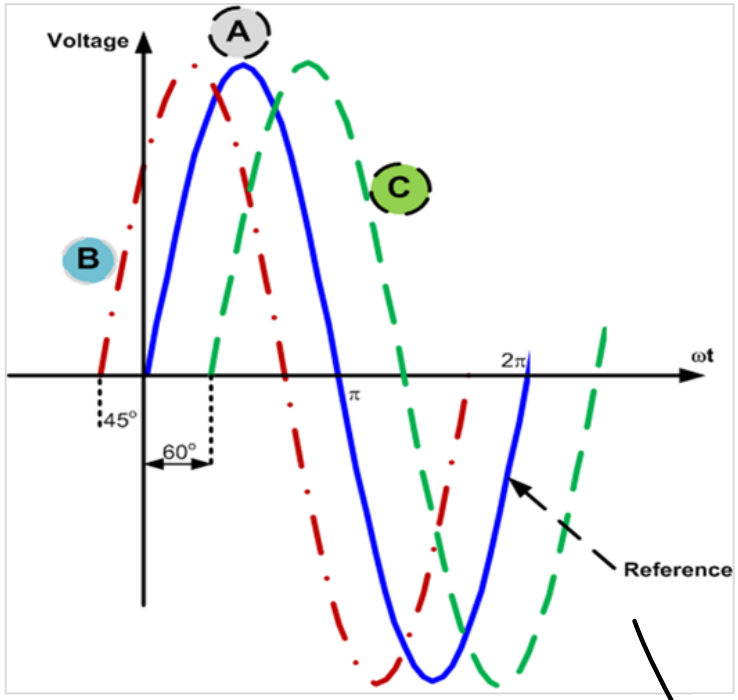




Notation: $v(t)$ - instantaneous voltage and
V is the corresponding Phasor voltage
V is a vector : Magnitude and phase

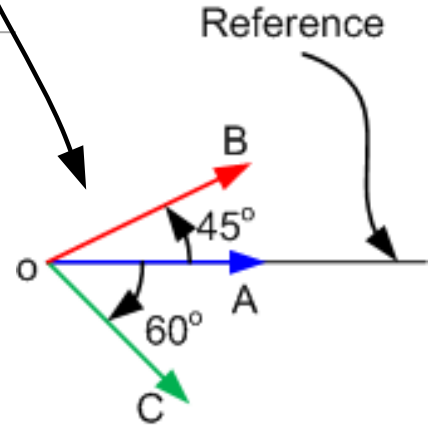
Phasor Representation

- Consider the following waveforms



- ✓ Waveform A is chosen as the reference and is shown to begin at 0°.
  $v(t) = V_m sin(\omega t)$; $\boldsymbol{V} = Vm\angle 0^o$
- ✓ Waveform B leads waveform A by 45 degrees
  $v(t) = V_m sin(\omega t + 45^o)$; $\boldsymbol{V} = Vm\angle 45^o$
- ✓ Waveform C lags waveform A by 60 degrees
  $v(t) = V_m sin(\omega t - 60^o)$; $\boldsymbol{V} = Vm\angle -60^o$

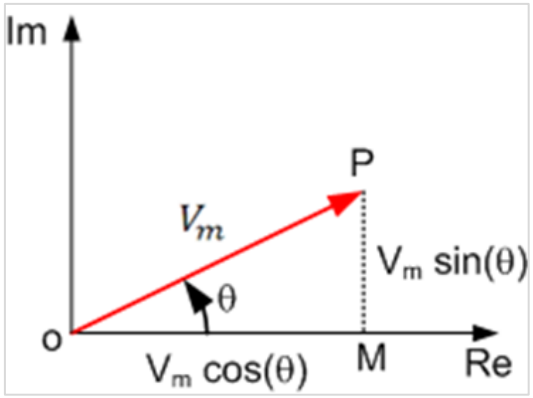
Phasor Representation



|OA| = |OB| = |OC|
Peak values of all three waveforms are equal

Phasors

- A phasor is a complex number that represents the amplitude and phase angle of a sinusoid.



If OP is a vector of magnitude V_m and the phase angle is θ degrees then, P can be represented in one of the following three forms:

- Rectangular: $OP = V_m \cos(\theta) + j\, V_m \sin(\theta)$
- Polar: $OP = V_m \angle \theta$
- Exponential: $OP = V_m e^{j\theta}$
- $V_m \angle \theta = V_m \cos(\theta) + j\, V_m \sin(\theta)$ {Polar to rectangular}

- Transforming a sinusoid to and from the time domain to the phasor domain

$$v(t) = V_m \sin(\omega t + \phi) \longleftrightarrow V = V_m \angle \phi$$

Time domain | Phasor Domain

Examples

Example 3

❑ Determine the phasors corresponding to the following sinusoids:

(a) $i(t) = 5\sin(10t + 10^o)$; (b) $v(t) = -7\cos(2t + 40^o)$;

Solution:

(a) $\mathbf{I} = 5\angle 10^o$

(b) $v(t) = 7\sin(2t - 90^o + 40^o)$ [we know that -cos ωt = sin (ωt-90)]

$\therefore \boldsymbol{V} = 7\angle -50^o$

Example 4

❑ Find the sinusoids represented by the following phasors:

(a) $\mathbf{V} = j8e^{-j20^o}$

Since $j = 1\angle 90^o$, $\boldsymbol{V} = j8\angle -20^o = (1\angle 90^o)(8\angle -20^o) = 8\angle 70^o$)

Converting this phasor to the time domain gives

$v(t) = 8\sin(\omega t + 70^o)$ Volts

(b) $\mathbf{I} = -3 + j4$

$Magnitude = \sqrt{(-3)^2+4^2} = 5$ and $\theta = tan^{-1}\frac{4}{-3} = 126.87^o$

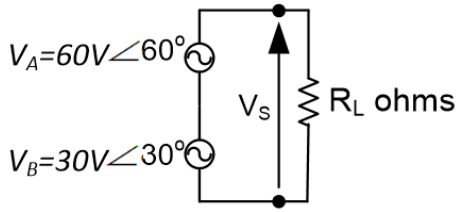
$\mathbf{I} = 5\angle 126.87^o$. Transforming this into time domain gives

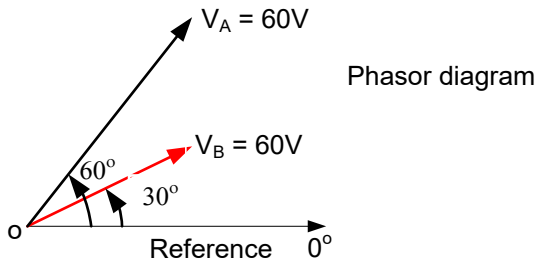
$i(t) = 5\sin(\omega t + 126.87^o)$ Amps

Examples

Example 5

❑ Two sine wave sources are connected in series across a load, as shown. Determine the total voltage, $V_S = V_A + V_B$.





Phasor diagram

Solution:

V_S=V_A+V_B = 60∠60° + 30∠30°

$V_m\angle\emptyset = V_m\cos(\emptyset) + j\,V_m\sin(\emptyset)$

V_S=[60 cos60^o + j 60 sin60^o] +[30 cos30^o + j 30 sin30^o]

V_S = 56 +j67

Magnitude = $\sqrt{56^2 + 67^2}$ = 87.3 V

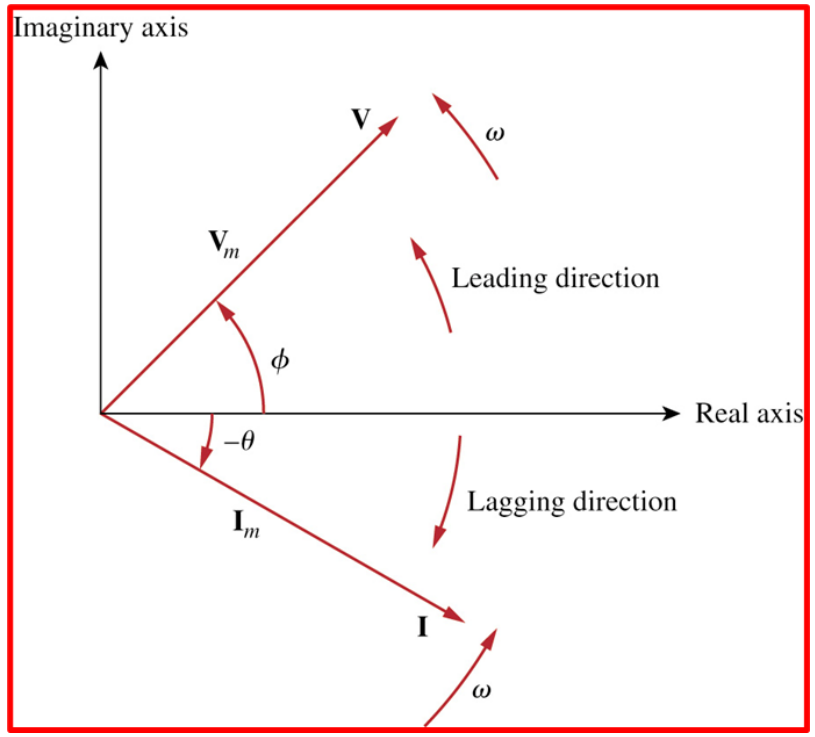
$\emptyset = tan^{-1}\frac{67}{56}$ = 50.2^o

Convert to polar form → V_S = 87.3V ∠50.2^o

Phase Difference

- Consider two phasors as given below and draw the phasor diagram.

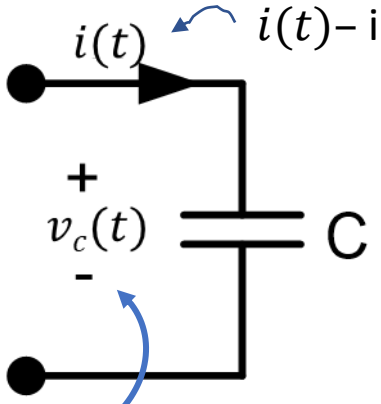
$V = V_m\angle\varphi$ and $I = I_m\angle\theta$



Phase difference between the Current and the voltage is: (φ-(-θ)) = (φ+θ)

Capacitors and Inductors

Capacitor



$i(t)$– instantaneous current

Assume that the initial charge on the capacitor is zero.

$v_c(t)$– instantaneous voltage across the capacitor

$q(t)$ - instantaneous charge on the capacitor

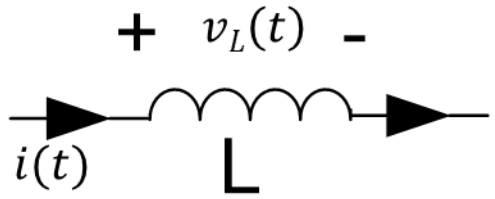
$$q(t) = Cv_c(t)$$

$$\frac{dq(t)}{dt} = C\frac{dv_c(t)}{dt}$$

$$i(t) = C\frac{dv_C(t)}{dt}$$

$$\therefore v_C(t) = \frac{1}{C}\int_0^t i(t)dt$$

Inductor



$v_L(t)$ – instantaneous voltage across the inductor

$i(t)$– instantaneous current

If $i(t)$ is the current through the coil, the voltage across the inductor $v_L(t)$ is given by

Induced voltage → $$v_L(t) = L\frac{di(t)}{dt}$$

The greater the rate of change of current through the coil, the greater will be the induced voltage.

The V–I relationship for a resistor, an inductor, and a capacitor

Resistor

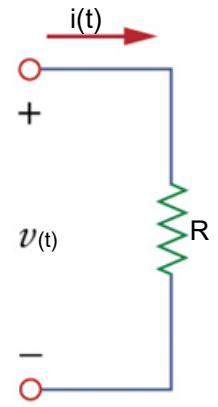
Assume that the current through the resistor is given by

$i(t) = I_m \sin(\omega t)$; **I** = $I_m \angle 0^o$ *(reference vector)*

Therefore $v(t) = i(t)\,R = R\,I_m \sin(\omega t)$

The phasor representation of the voltage is :

$$\boldsymbol{V} = RIm\angle 0^o$$



Phasor diagram



The phase angle of both voltage and the current waveforms is 0°. Both **I** and **V** are said to be in phase.

Inductor

Assume that the current through the inductor is given by

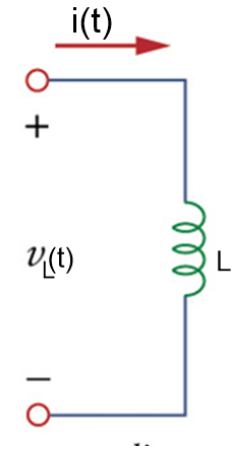
$i(t) = I_m \sin(\omega t)$; $\boldsymbol{I} = Im\angle 0^o$ *(reference vector)*

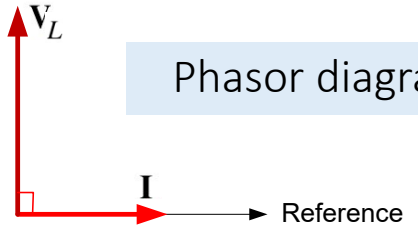
We have $v_L(t) = L\frac{di(t)}{dt} = \omega L I_m \cos(\omega t)$

$$v_L(t) = \omega L I_m \sin(\omega t + 90^o)$$

The phasor representation of the voltage is :

$$\boldsymbol{V_L} = \omega\, L\, I_m \angle 90^o$$





Phasor diagram

The voltage leads the current by 90° **OR** the current lags the voltage by 90° .

The V–I relationship for a resistor, an inductor, and a capacitor

Capacitor

Assume that the Voltage across the capacitor is given by

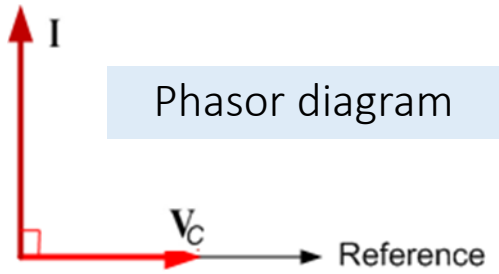
$v_c(t) = V_m \sin(\omega t)$; $\mathbf{V_C} = V_m \angle 0^o$ *(reference vector)*

We have $i(t) = C\frac{dv_c(t)}{dt} = \omega C V_m \cos(\omega t)$

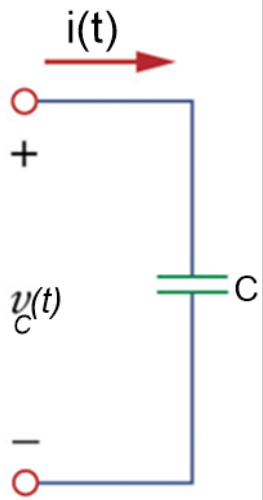
$i(t) = \omega C V_m \sin(\omega t + 90^o)$

The phasor representation of the current is :

$\boldsymbol{I} = \omega C V_m \angle 90^o$



Phasor diagram

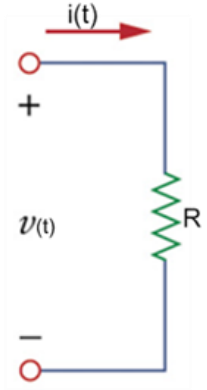


The current leads the voltage by 90° **OR** the voltage lags the current by 90°

Note: The impedance (Z) of a circuit element in an AC circuit is the ratio of the voltage phasor to the current phasor.

Impedance of Circuit Elements

Resistor:



$\mathbf{I} = I_m \angle 0^o$

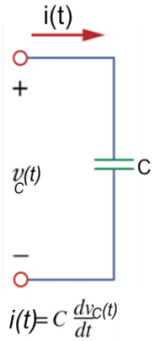
$\mathbf{V} = RI_m \angle 0^o$

$\mathbf{V} = RI_m$

$\mathbf{V} = R\,\mathbf{I}$

$Impedance\ z = \frac{V}{\mathbf{I}} = R$

Capacitor:



$\mathbf{V_c} = V_m \angle 0^o$

$\mathbf{I} = \omega C V_m \angle 90^o$

$\mathbf{I} = j\omega C\ V_m$

$\mathbf{V_c} = \frac{1}{j\omega C}\mathbf{I}$

Impedance $\mathbf{z_L} = \frac{\mathbf{V_L}}{\mathbf{I}} = j\omega L$

Inductor:

$\mathbf{I} = I_m \angle 0^o$

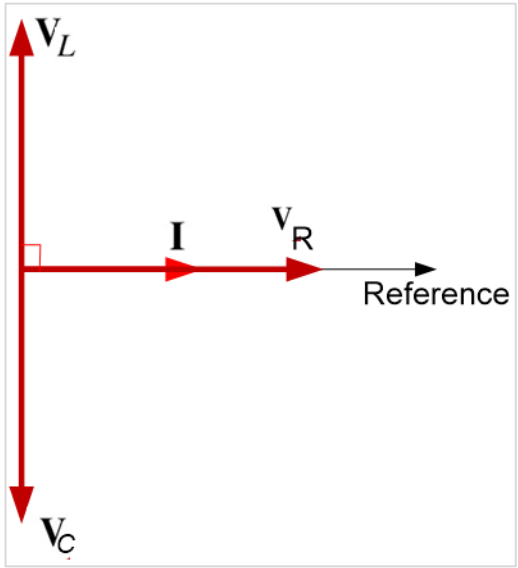
$V_L = \omega L I_m \angle 90^o$

$V_L = j\omega L\ I_m$

$\mathbf{V_L} = j\omega L\ \mathbf{I}$

$Impedance\ \mathbf{zC} = \frac{\mathbf{V_c}}{\mathbf{I}} = \frac{1}{j\omega C}$

Phasor relationships for Circuit Elements



Summary:

- The impedance (Z)of a resisitor is R,
- The impedance of an inductor is jωL
- The impedance of a capacitor is 1/jωC

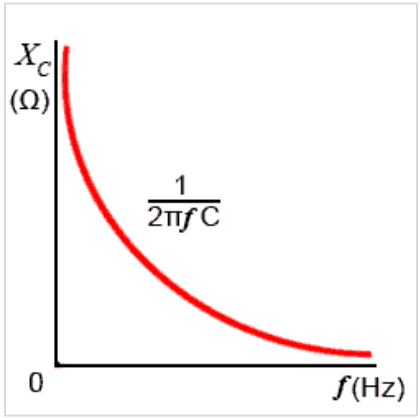
Reactance - The imaginary part of the impedance is called reactance

Capacitor

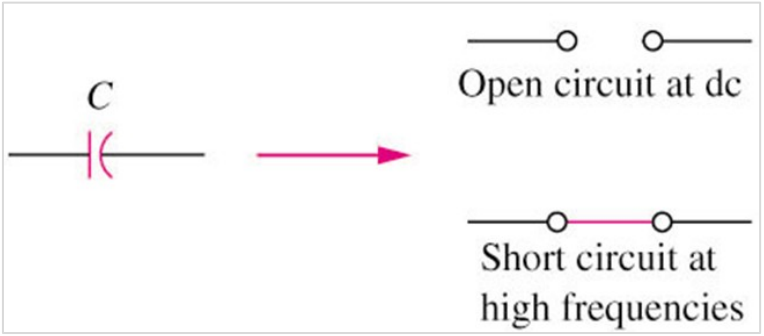
The impedance of a capacitor $(Z_c) = \frac{1}{j\omega C}$

The capacitive Reactance (like resistance) $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ *ohms*

$$Z_C = \frac{1}{j\omega C} = -jX_C$$



As frequency increases the capacitive reactance decreases



$\omega = 0; Z \to \infty$

$\omega \to \infty; Z = 0$

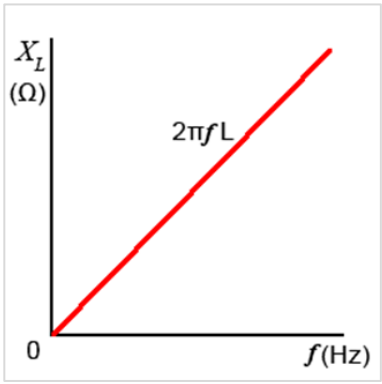
Inductor

The impedance of a inductor $(Z_L) = j\omega L$

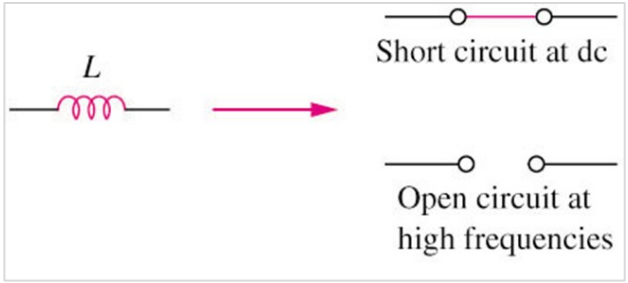
The inductive Reactance (like resistance):

$X_L = \omega L = 2\pi f L$ *ohms*

$Z_L = j\omega L = jX_L$



As frequency decreases the inductive reactance increases

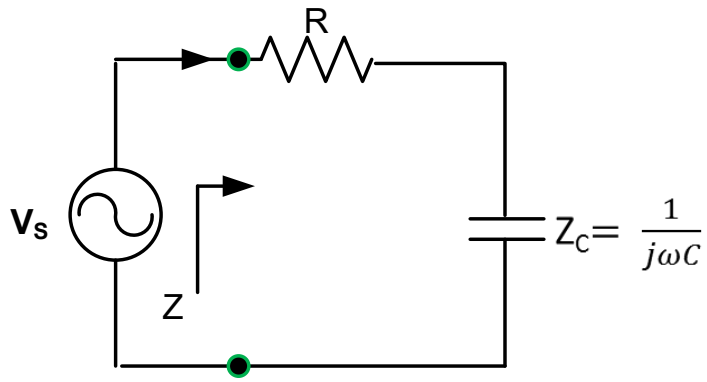


$\omega = 0; Z = 0$

$\omega \to \infty; Z \to \infty$

Impedance in a RC circuit

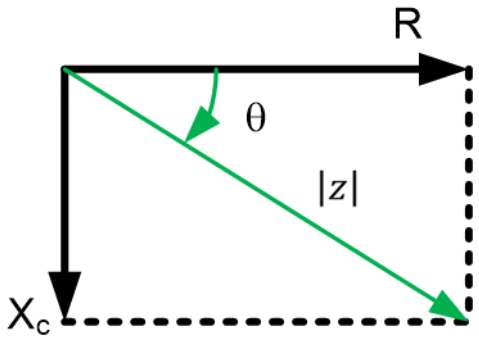
- Impedance in a RC circuit is composed of resistance and reactance.



$$Impedance\ z = R + \frac{1}{j\omega c} = R - \frac{j}{\omega c} = R - jX_c$$

$$|z| = \sqrt{R^2 + \left(\frac{-1}{\omega c}\right)^2} = \sqrt{R^2 + {X_c}^2}$$

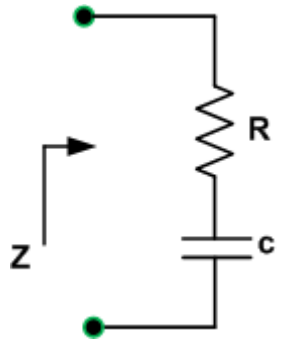
$$\theta = tan^{-1}\left(\frac{-1}{\omega cR}\right) = tan^{-1}\left(\frac{-X_c}{R}\right)$$



Impedance Triangle

Example 6

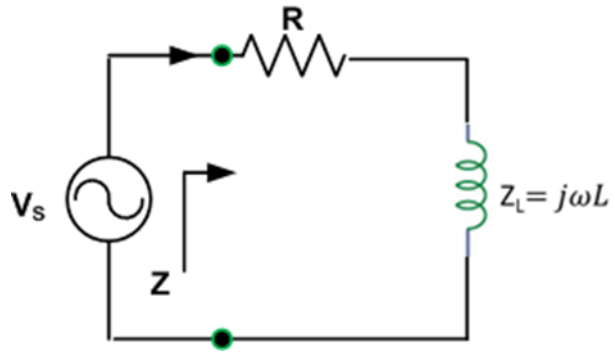
- A capacitive reactance of 2.4 kΩ is in series with a resistance of 1.2 kΩ Calculate the impedance.



$$z = R - jXc$$
$$z = 1.2 - j\ 2.4$$
$$z = 2.63\ \text{k}\Omega \quad \angle -63.4^o$$

Impedance in a RL circuit

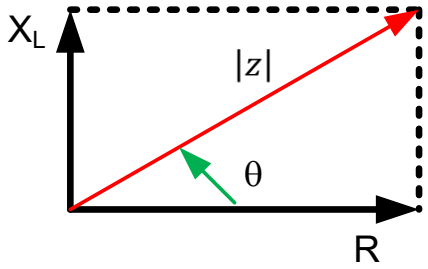
- Impedance in a RL circuit is composed of resistance and reactance.



$Impedance\ z = R + j\omega L$ = R + jX_L

$|z| = \sqrt{R^2 + (\omega L)^2}$ = $\sqrt{R^2 + X_L{}^2}$

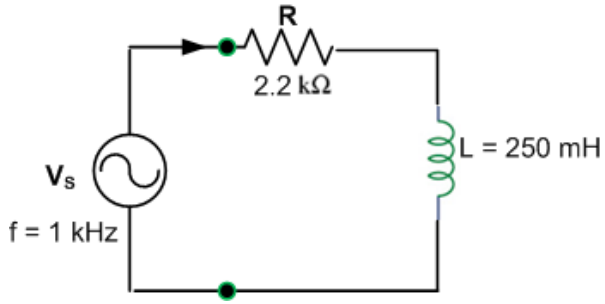
$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$ = $\tan^{-1}\left(\frac{X_L}{R}\right)$



Impedance Triangle

Example 7

- For the circuit shown below, calculate the value of the impedance.



$X_L = 2\pi f L = 2\pi\ 1000\ \ 250\ 10^{-3}$ = 1.5 kΩ

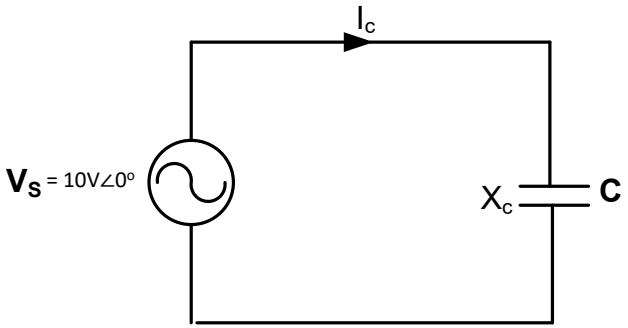
Impedance $z\ = \ R + \ jX_L$ = 2.2 + j 1.57

$z = 2.7k\Omega\ \angle 35.5^o$

Exercises

Example 8

- A sine wave source of 10V∠0° is applied across a 0.22μF capacitor. If the frequency of the source is 2kHz, compute the current.



$$X_c = \frac{1}{2\pi fc} = \frac{1}{2\pi\,(2000)(0.22\,10^{-6})} = 3.619\ k\Omega$$

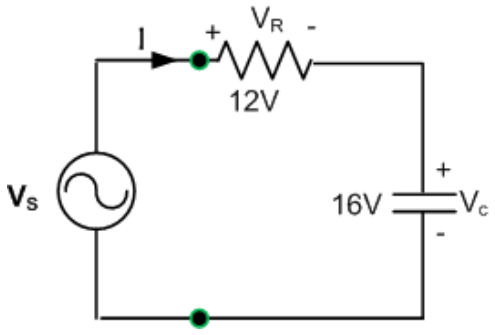
$$Z_c = \frac{1}{j2\pi fc} = 3.619\ k\Omega\ \angle\text{-}90^o$$

$$I_c = \frac{V_s\angle 0^o}{X_c\angle -90^o} = \frac{10\text{ V}\ \angle 0^o}{3.619\ k\Omega\ \angle -90^o} = 2.76\text{ mA}\ \angle 90^o$$

I_c leads the input voltage V_s by 90°

Example 9

- An AC source across an RC circuit produces a V_R of 12V and a V_C of 16V. Compute the source voltage.

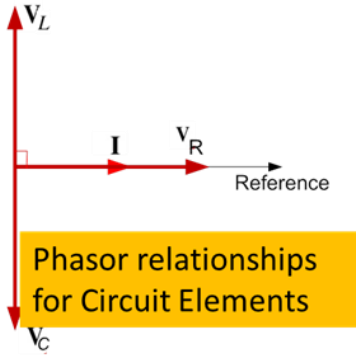


$$V_s = VR - jVc = 12 - j16$$

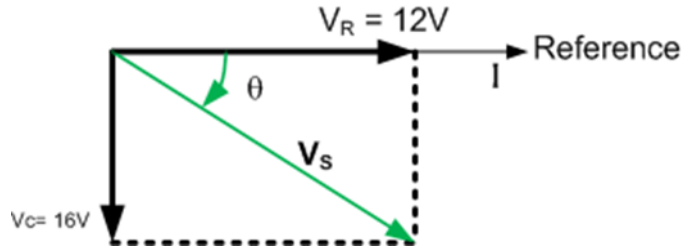
$$|V_s| = \sqrt{12^2 + 16^2} = 20\text{V}$$

$$\theta = tan^{-1}\frac{-16}{12} = \text{-}53.1^o$$

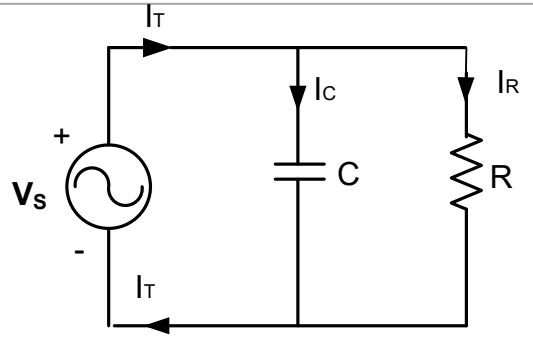
$$V_s = 20V\angle\text{-}53.1^o$$



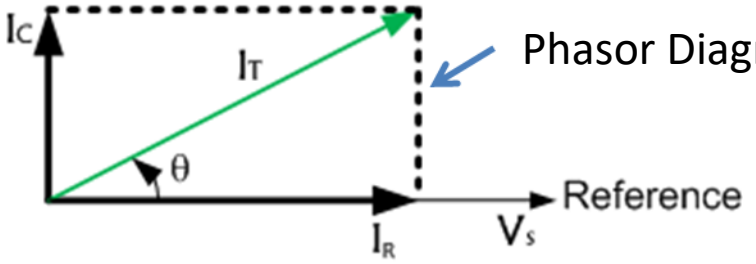
Phasor relationships for Circuit Elements



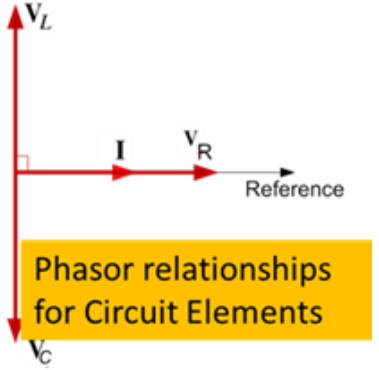
Analysis of parallel RC circuits



- In parallel circuits the supply voltage is taken as the reference.
- In series circuits the current is taken as the reference.



Phasor Diagram



Phasor relationships for Circuit Elements

From the phasor diagram, we can write the following equations:

$$\boldsymbol{I}_T = \boldsymbol{I}_R + j\,\boldsymbol{I}_C$$

$$|I_T| = \sqrt{I_R{}^2 + I_C{}^2}$$

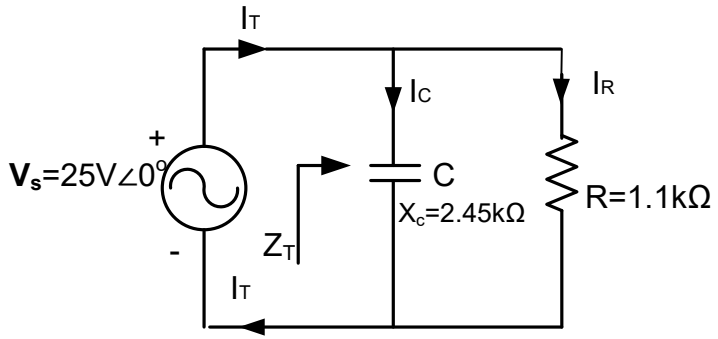
$$\tan\theta = \frac{\boldsymbol{I}_C}{\boldsymbol{I}_R}$$

$$\boldsymbol{I}_T = \sqrt{I_R{}^2 + I_C{}^2}\,\angle\theta^{\circ}$$

Examples

Example 10

- Compute the total current for the following circuit



Method 1:

$$\mathbf{Z}_c = \frac{1}{j\omega c} = 2.45\text{k}\boldsymbol{\Omega} \angle -90^\circ$$

$$\mathbf{I}_c = \frac{V_s}{Z_c} = \frac{25\text{V}\angle 0^\circ}{2.45\text{k}\boldsymbol{\Omega} \angle -90^\circ} = 10.2 \text{ mA} \angle 90^\circ$$

$$\mathbf{I}_R = \frac{V_s}{R} = \frac{25\text{V}\angle 0^\circ}{1.1\text{k}\boldsymbol{\Omega} \angle 0^\circ} = 22.7 \text{ mA} \angle 0^\circ$$

$$\mathbf{I}_T = \mathbf{I}_c + \mathbf{I}_R = 10.2 \text{ mA} \angle 90^\circ + 22.7 \text{ mA} \angle 0^\circ$$

$$\mathbf{I}_T = 22.7 + \text{j } 10.2 \text{ mA}$$

$$\mathbf{I}_T = 24.9 \text{ mA} \angle 24.2^\circ$$

Method 2:

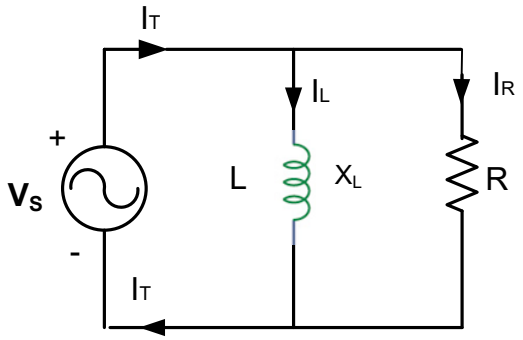
Total impedance of the circuit is given by

$Z_T = R//(1/j\omega c)$

$$Z_T = \frac{R.\frac{1}{j\omega c}}{R+\frac{1}{j\omega c}} = \frac{1.1\text{ k}\boldsymbol{\Omega}}{1+j\,(2.45\text{k}\boldsymbol{\Omega})\,(1.1\text{k}\boldsymbol{\Omega})}$$ Z_T =1.04∠-24.2°

$$\mathbf{I}_T = \frac{25\text{V}\angle 0^\circ}{1.04\angle -24.2^\circ} = 24.9 \text{ mA} \angle 24.2^\circ$$

Analysis of parallel RL circuits



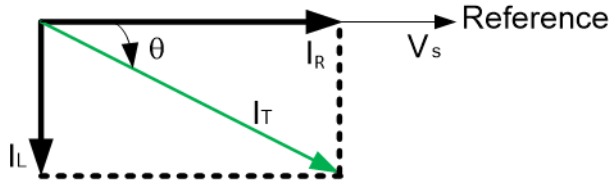
Using the Phasor Diagram

We can write the following equations:

$$\boldsymbol{I_T = I_R + I_L}$$

$$|IT| = \sqrt{I_R{}^2 + I_L{}^2}$$

$$\tan\theta = \frac{\boldsymbol{I_L}}{\boldsymbol{I_R}}$$



$$\boldsymbol{I_T} = \sqrt{I_R{}^2 + I_L{}^2}\ \angle -\theta^o$$

Using Complex numbers

Total impedance of the circuit is given by

$$Z_T = R//(jX_L) = \frac{R.jX_L}{R+jX_L} = |z|\angle\beta^o$$

$$I_T = \frac{V_s\angle 0^o}{|z|\angle\beta^o}$$

OR

$$I_L = \frac{V_s\angle 0^o}{jX_L} = \frac{V_s\angle -90^o}{X_L}$$

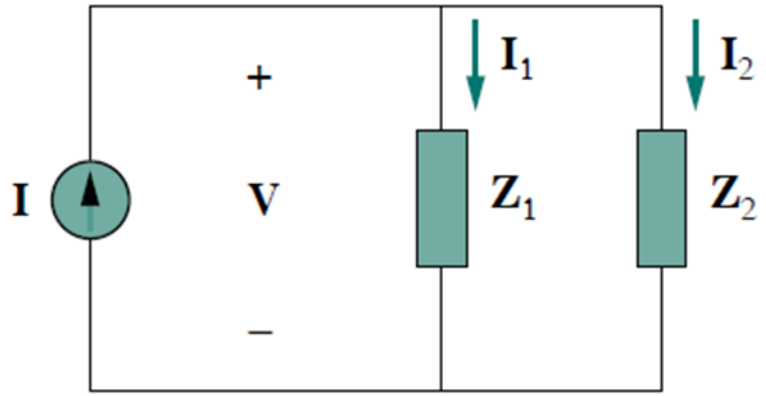
$$I_R = \frac{V_s\angle 0^o}{R}$$

$$\boldsymbol{I_T = I_R + I_L}$$

Current and Voltage Division

Current Division

Ideal current source

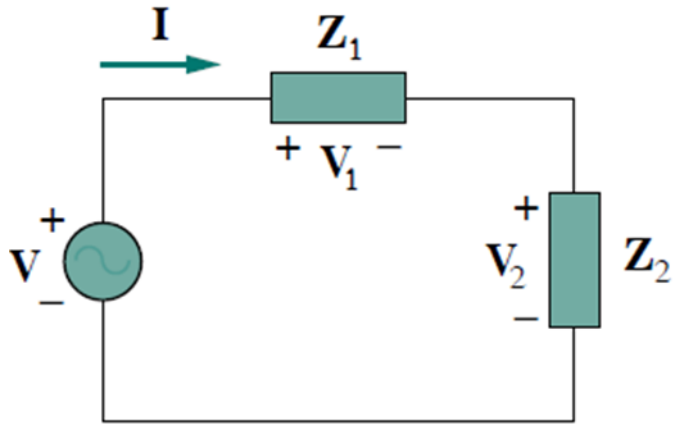


$$\mathbf{I}_1 = \frac{Z_2}{Z_1 + Z_2}\mathbf{I}$$

$$\mathbf{I}_2 = \frac{Z_1}{Z_1 + Z_2}\mathbf{I}$$

Voltage Division

Ideal voltage source



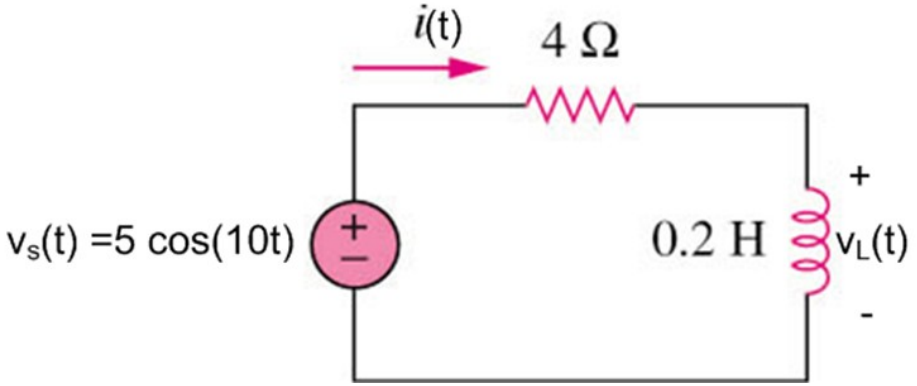
$$\mathbf{V}_1 = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}$$

$$\mathbf{V}_2 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}$$

Examples

Example 11

❑ Determine the current $i(t)$ and $v_L(t)$ across the inductor.



$$\omega = 10 \text{ rad/s}$$

$$\mathbf{V}_s = 5\angle 0^\circ, \quad X = j\omega L = j2 = 2\angle 90^\circ$$

$$Z = R + j\omega L = 4 + j(10 \times 0.2) = 4 + j2 = 4.5\angle 26.6^\circ$$

$$\mathbf{I} = \frac{\mathbf{V}_s}{Z} = 1.11\angle -26.6^\circ \qquad i(t) = 1.11\cos(10t - 26.6^\circ)$$

$$\mathbf{V} = I(j\omega L) = (1.11\angle -26.6^\circ) \times (2\angle 90^\circ) = 2.22\angle 63.4^\circ$$

$$v(t) = 2.22\cos(10t + 63.4^\circ)$$

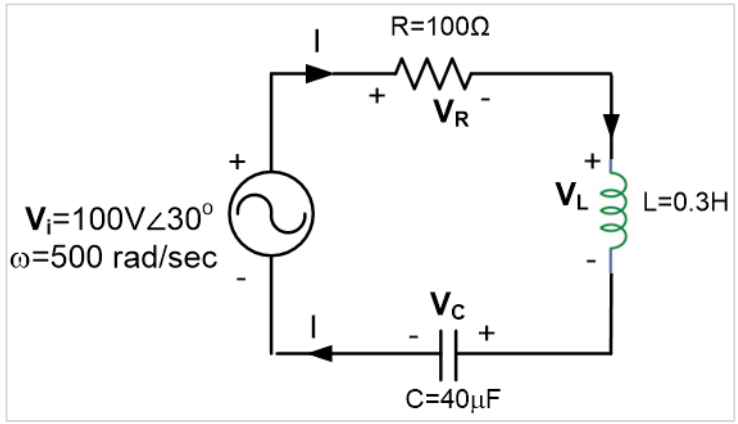
Alternatively, by voltage division:

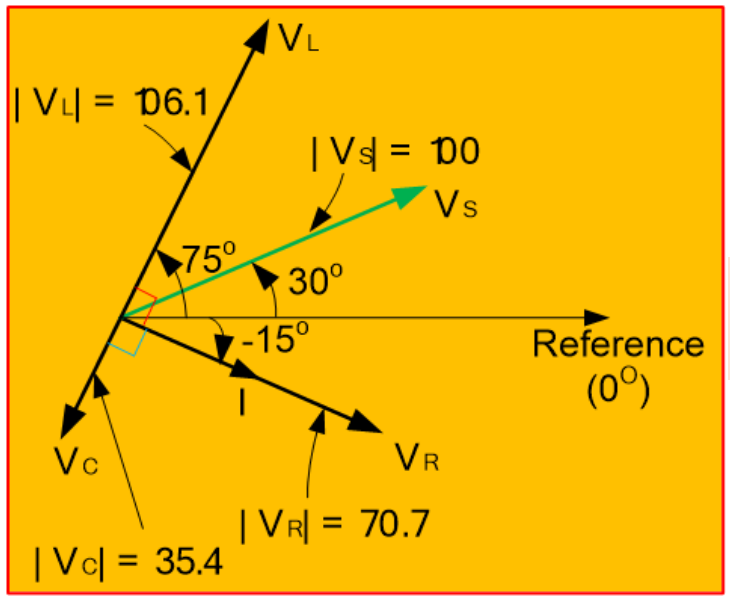
$$\mathbf{V} = \frac{j2}{4 + j2} \times 5\angle 0^\circ = \frac{10\angle 90^\circ}{4.47\angle 26.56^\circ} = 2.2\angle 63.4^\circ$$

Example - RLC circuit

Example 12

- Find the steady-state current for the circuit shown below. Find the phasor voltage across each element and construct a phasor diagram.





Phasor Diagram

$\mathbf{V}_i = 100\angle 30^o$

$Z_L = j\omega L = j(500)\,(0.3) = j150\ \Omega$

$Z_C = (1/j\omega c) = \dfrac{-j}{(500)(40 \times 10^{-6})} = -j50\Omega$

The total impedance:

$Z_T = R + Z_L + Z_C = 100 + j150 - j50 = 100 + j100$

$Z_T = 141.4\angle 45^o$

The Current ***I***

$\mathbf{I} = \dfrac{V_i}{Z_T} = \dfrac{100\angle 30^o}{141.4\angle 45^o} = 0.707\angle -15^o$

$\mathbf{V}_R = R\,\mathbf{I} = (100)(0.707\angle -15^o) = 70.7\angle -15^o$

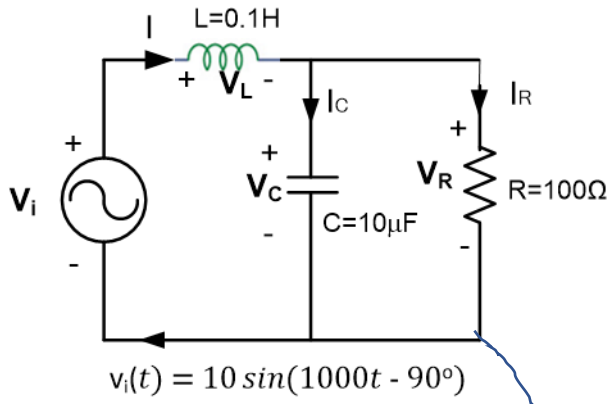
$\mathbf{V}_L = j\omega L\,\mathbf{I} = (150\angle 90^o)\,(0.707\angle -15^o) = 106.1\angle 75^o$

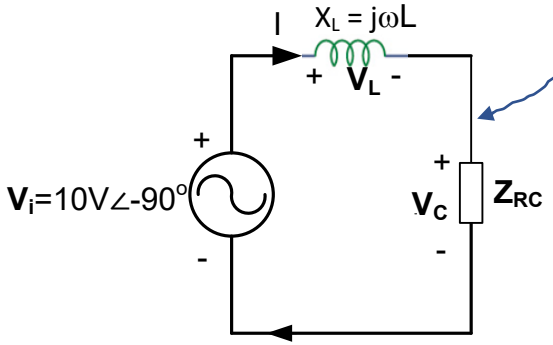
$\mathbf{V}_C = (1/jwc)\mathbf{I} = (-j/\omega c)\mathbf{I} = (50\angle -90^o)\,(0.707\angle -15^o)$
$= 35.4\angle -105^o$

Example - Series and Parallel combinations of complex impedances

Example 13

- For the circuit shown below, Find the current through each element and construct a phasor diagram showing the currents and the source voltage.





Solution

$\mathbf{V}_i = 10\angle -90°$ and ω = 1000 rad/sec

$Z_L = j\omega L = j(1000)\,(0.1) = j100\ \Omega$

$Z_C = (1/j\omega c) = \dfrac{-j}{(1000)(10 \times 10^{-6})} = -j100\Omega$

$Z_{RC} = \dfrac{1}{\left(\frac{1}{R}\right)+\left(\frac{1}{Z_c}\right)} = \dfrac{1}{0.01 + j\,0.01} = 70.71\angle -45°$

$Z_{RC} = 50 - j50$

$\mathbf{V}_C = \dfrac{Z_{RC}}{(Z_L)+(Z_{RC})}\mathbf{V}i = \dfrac{(70.71\angle -45°)\,10\angle -90°}{j100+(50-j50)} = 10\angle -180°$

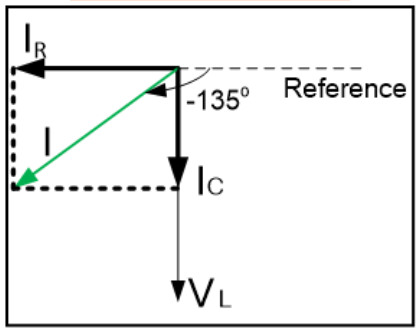
Now we compute the current in each element

$\mathbf{I} = \dfrac{\boldsymbol{V}_i}{Z_L+ZRC} = \dfrac{10\angle -90°}{j100+(50-j50)} = 0.1414\angle -135°$

$\mathbf{I}_R = \dfrac{\boldsymbol{V}_C}{R} = \dfrac{10\angle -180°}{100} = 0.1\angle -180°$

$\mathbf{I}_C = \dfrac{\boldsymbol{V}_C}{Z_C} = \dfrac{10\angle -180°}{-j100} = 0.1\angle -90°$

Phasor Diagram



ELEC2134: Circuits and Signals

AC Circuit Analysis

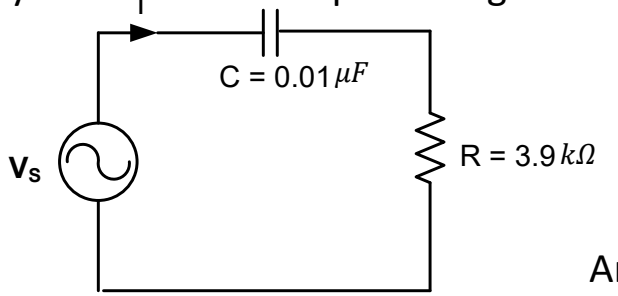
Tutorial questions

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

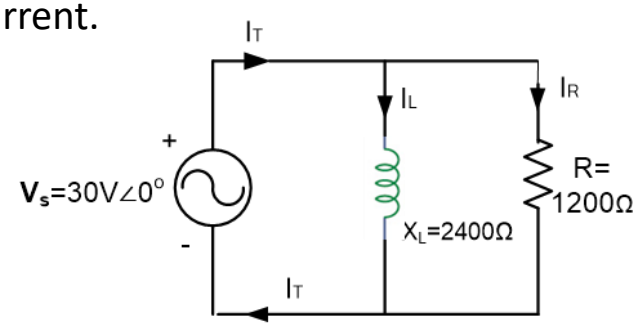
Tutorial Questions (AC Circuit Analysis)

Question 1: The circuit below has a phase angle of 36°. What frequency would cause the phase angle to double to 72°?



Ans: 1.326 kHz

Question 2: For the parallel RL circuit, calculate the total current.



Ans: 28mA ∠−26.6°

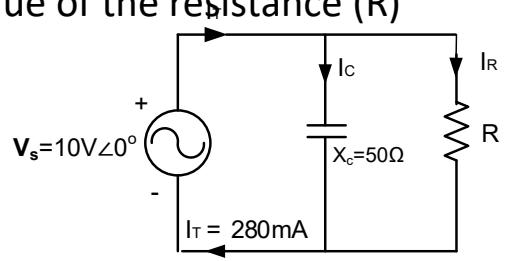
Question 3: A circuit element is subjected to the current $i(t) = 10\cos(2\pi 103t + 120^o)$ mA. Derive an expression for the voltage waveform as well as its phasor if the element is (i) a 2k **Ω** resistance. (ii) a 0.5 μF capacitance and (iii) a 10mH inductance.

Ans: 20∠120°, 3.183∠30°, 0.629∠-150°

Question 4: Given $i_1(t) = 4\cos(\omega t + 30^\circ)$ A and $i_2(t) = 5\sin(\omega t - 20^\circ)$ A, find their sum $i(t) = i_1(t) + i_2(t)$.

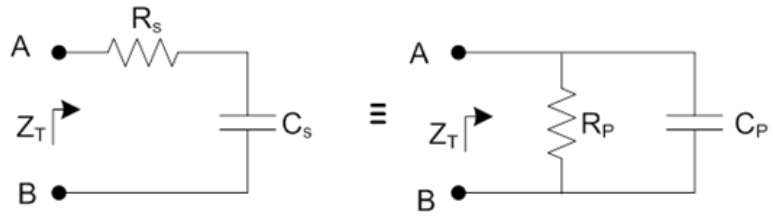
Ans: $3.218 \angle 33^\circ$ A

Question 5: For the circuit shown, compute the value of the resistance (R)



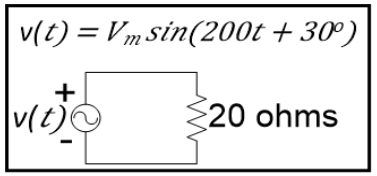
Ans: 51 ohms

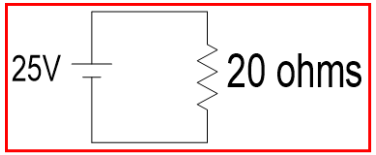
Question 6: The following two circuits are equivalent. Derive an expression for each R_P and C_P in terms of R_s, C_s & ω.



Ans: $R_s = \frac{R_p}{1+R_p^2C_p^2\omega^2}$; $C_s = \frac{1+R_p^2C_p^2\omega^2}{R_p^2C_p\omega^2}$

Question 7: The sources in the AC and DC circuits shown below supply equal values of average power. Determine the values of average power and of the amplitude of the sinusoidal voltage.





Ans: P_{ave} = 31.25 W, V_m = 35.355 V

Question 8: Derive an expression for the rms value of a triangular voltage waveform given below:

